CMM Exam Sheet 2025/26 - Non ii, non vidi, non vicī

Thm: (Stokes). Let Σ be a smooth oriented surface in $\mathbb{R}^3$ with boundary $\partial\Sigma \equiv \Gamma$. If a vector field $F(x,y,z) = (F_x(x,y,z), F_Y(x,y,z), F_z(x,y,z))$ is defined and has cont. 1^{st} order partial derivatives in a region containing Σ, then $\iint_\Sigma (\nabla \times F)\cdot d\Sigma = \oint_{\partial\sigma} F \cdot d\Gamma$. More explicitly, the equality says that $\iint_\Sigma\left(\left(\frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z}\right)dydz + \left(\frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x}\right)dzdx + \left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)dxdy\right) = \oint_{\partial\Sigma}(F_x dx + F_y dy + F_z dz)$.

Dfn: (SVCDE Identities). $\boldsymbol{a}\times\boldsymbol{b} = \|\boldsymbol{a}\|\|\boldsymbol{b}\|\sin(\theta)\boldsymbol{n}$, $\boldsymbol{n}$ norm to $\boldsymbol{ab}$-plane

- $\nabla\cdot(\nabla\times F) = 0$
- $\mathbf{x}\times\mathbf{y} = -(\mathbf{y}\times\mathbf{x})$
- $\nabla(|\mathbf{x}|) = \frac{\mathbf{x}}{|\mathbf{x}|}$
- $\nabla\times(\nabla F) = 0$
- $\mathbf{x}\cdot\mathbf{y} \le \|\mathbf{x}\|\|\mathbf{y}\|$
- $\mathbf{x}\times\mathbf{x} = 0$
- $\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\| + \|\mathbf{y}\|$
- $\nabla(\frac{k}{|\mathbf{x}|}) = -\frac{k\mathbf{x}}{|\mathbf{x}|^3}$

Thm: For C a curve $P \to Q$, $\int_C \nabla F\cdot ds = F(Q) - F(P)$

1. Newtonian Mechanics

Dfn: The universe, to a good approx, looks like $\mathbb{R}\times\mathbb{R}^3$, where the 1^{st} factor $\mathbb{R}$ represents **time**, and the 2^{nd} factor $\mathbb{R}^3$ represents **space**.

Dfn: A point $(t,\mathbf{a}) \in \mathbb{R}\times\mathbb{R}^3$ is called an **event**. Two events $(t,\mathbf{a})$ and $(t',\mathbf{b})$ are **simultaneous** iff $t = t'$. The distance between simultaneous events is given by the norm $|\mathbf{a}-\mathbf{b}|$.

Dfn: A particle's trajectory through the universe is given by the **worldline**. This is a graph $\mathbf{x}:\mathbb{R}\to\mathbb{R}^3, t\mapsto\mathbf{x}(t)$,of form$\{(t,\mathbf{x}(t))\in\mathbb{R}\times\mathbb{R}^4\}$, in the Newtian universe.

Dfn: The **configuration space** of a many body system is given by the locations of each particle in space, so $\mathbb{R}^3\times\mathbb{R}^3\times\cdots\times\mathbb{R}^3 = \mathbb{R}^N$, $N\equiv 3n$. The **state space** of such systems is given by the collection of positions and velocities of each of the particles $(\mathbf{x},\dot{\mathbf{x}})$. Thus has dim $=6n$.

Dfn: For $\mathbf{x}_a(t):\mathbb{R}\to\mathbb{R}^3$ a worldline of particle a, the collection of these worldlines defines a curve $\mathbf{x}(t) = (\mathbf{x}_1(t),\mathbf{x}_2(t),\ldots,\mathbf{x}_n(t))$, in the configuration space.

Princ: (Newt's Principle of Determinancy). The motion of any mechanical system is uniquely determined by its initial state, the set of all its constituent positions and velocities at a given instant in time. Equivalently the worldline $\mathbf{x}(t)$ in the configuration space is determined by $\mathbf{x}(0)$ and $\dot{\mathbf{x}}(0)$ for all $t\in\mathbb{R}$ or at least for all $t\in[0,\tau)$ for some $\tau>0$.

Dfn: (Newt's Eqn). By Newt's PoD there must be a relationship of form $\ddot{\mathbf{x}}(t) = \Phi(\mathbf{x},\dot{\mathbf{x}},t)$ for some $\Phi:\mathbb{R}^N\times\mathbb{R}^N\times\mathbb{R}\to\mathbb{R}^N$. Let $v:\mathbb{R}\to\mathbb{R}^N$ then the system $\begin{pmatrix}\dot{\mathbf{x}}(t)\\ v(t)\end{pmatrix} = \begin{pmatrix}v(t)\\ \Phi(\mathbf{x},v,t)\end{pmatrix}$ has the same soln space as Newt's eqn.

Princ: (Newt's 1^{st} Law). A particle (or many-body system of particles) is freely-moving if $\Phi(\mathbf{x},\dot{\mathbf{x}},t) = 0$. In this case, Newt's eqn iss $\ddot{\mathbf{x}}(t) = 0$. If the initial state of such a freely moving system is $\mathbf{x}(0) = \mathbf{x}_0$, $\dot{\mathbf{x}}(0) = v_0$, then the corresponding Newt's eqn is solved by $\mathbf{x}(t) = \mathbf{x}_0 + tv_0$. Thus the freely moving body moves on a straight worldline at constant vel..

Princ: (Newt's 2^{nd} Law). A force is a map $F:\mathbb{R}^3\to\mathbb{R}^3$ which affects the vel. of a particle. For a particle of (inertial) mass m, the Newt's eqn in a force field becomes $F(\mathbf{x}) = m\ddot{\mathbf{x}}$,or$\Phi(\mathbf{x},\dot{\mathbf{x}},t) = \frac{F(\mathbf{x})}{m}$.

Dfn: A point $\mathbf{x}\in\mathbb{R}^3$ for which $F(\mathbf{x}) = 0$ is an **equil. point** of the force field F.

2. Galilean Relativity

Dfn: A **Galilean transform** is an affine transform which preserves time intervals and the distances between simul. events.

Lem: A Galil. transform is a linear transform of form $\begin{pmatrix}1&0&s\\ v&R&s\\ 0&0&1\end{pmatrix}$ with $s,v\in\mathbb{R}^3$, $R:\mathbb{R}^3\to\mathbb{R}^3$ an orthog transform. Consider three types:

Type 1: $\begin{pmatrix}1&0&s\\0&1&1\\0&0&1\end{pmatrix}\begin{pmatrix}t\\x\\1\end{pmatrix} = \begin{pmatrix}t+s\\x+s\\1\end{pmatrix}$ Type 2: $\begin{pmatrix}1&0&0\\v&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}t\\x\\1\end{pmatrix} = \begin{pmatrix}t\\x+vt\\1\end{pmatrix}$ Type 3: $\begin{pmatrix}1&0&0\\0&R&0\\0&0&1\end{pmatrix}\begin{pmatrix}t\\x\\1\end{pmatrix} = \begin{pmatrix}t\\Rx\\1\end{pmatrix}$ Type 1 is translation/change of origin in $\mathbb{R}^4$. Type 2 is motion with fixed, uniform vel. v. Type 3 is re-orientation of spatial axes.

Princ: (Galilean Relativity). The laws of motion (Newt's Eqn) are invariant under Galilean transforms - translation, fixed $\dot{x}$ motion and rotation of spatial axes.

3. Conservation Laws

Dfn: A **conserved qtty** is a func of $(\mathbf{x},v)$ which remains constant in time - ie has time-deriv 0 along solns of the eqns of motion.

Dfn: $F:\mathbb{R}^3\to\mathbb{R}^3$ is **conservative** if it can be written as $F = -\nabla U$ for some func $U:\mathbb{R}^3\to\mathbb{R}$ where ∇ is the gradient operator.

Dfn: The **potential** of a conservative force field F is $U(\mathbf{x})$. Potential is a func $U:\mathbb{R}^N\to\mathbb{R}$ s.t. $\ddot{\mathbf{x}} = -\nabla U$.

Dfn: The **energy** of a particle of mass m is $E(\mathbf{x},v) = \frac{m}{2}|v|^2 + U(\mathbf{x})$ where $T(v) = \frac{m}{2}|v|^2$ is kinetic energy and $U(\mathbf{x})$ is potential energy. $E = T+U$.

Lem: In a conservative force field, the energy is a cnsrvd qtty.

Dfn: Along some path $C\subset\mathbb{R}^3$ with endpoints x_0 and x_1, the **Work** W, done by a force field F from x_0 to x_1 via the path C is $W = \int_{x_0}^{x_1} F\cdot dx = \int_C F\cdot dx$. Paramise by $x(t)$ s.t. $x_0 = x(t_0)$, $x_1 = x(t_1)$ then $W = \int_{t_0}^{t_1} F\cdot vdt$.

Thm: Let F be a force field, then TFAE.

1. F is conservative, $F = -\nabla U$ for some $U:\mathbb{R}^3\to\mathbb{R}$.
2. The work of F along any path C connecting x_0 to x_1 depends only on x_0,x_1 and not on the shape of C, and if C is a closed path then the work of F along C is 0.
3. The curl of F vanishes, $\nabla\times F = 0$.

Dfn: Motion in 1D for a conservative force is governed by $m\ddot{x} = -\frac{dU}{dx}$. If $U(x)$ is linear then motion is the same as the example of Galil. gravity. When $U(x)$ is quadratic — this is "simple harmonic motion".

Princ: (Hooke's Law). The restoring force of a spring is well approxed by the potential $U(x) = \frac{k}{2}x^2$ where $k>0$ is a constant characterising stiffness of the spring. The resulting force by Hooke's Law is $F = -kx$ and Newt's eqn becomes $m\ddot{x} = -kx$. The amplitude of the spring's oscillation is $A = \sqrt{2E/k}$. Physical trajectories in the state space $\mathbb{R}^2$.

Lem: In any 1D conservative system, $E = \frac{m}{2}v^2 + U(x)$, is a constant of motion. As $\frac{m}{2}v^2 \ge 0$, it follows that $E \ge U(x)$ with equality iff $v = 0$. Configurations with $E < U(x)$ are not allowed.

With the system at fixed energy E, there are two permitted regions of motion: for $x(t)\in[a,b]$ the motion will be oscillatory (in fact periodic); and for $x(t)\in[c,\infty)$ motion will tend towards $x\to\infty$. In the oscillatory case the period of motion is $T = \sqrt{2m}\int_a^b 1/\sqrt{E-U(x)}$ dx. This integral converges iff a and b are not CPs of the potential. If b is such a CP the particle takes an 'infinite' amount of time to reach b.

Now suppose the energy is increased until it coincides with a local max of the potential as shown. If the particle starts from rest at $x(0) = a$, the previous formula suggests periodic motion between a and b but this can't happen now as b is an equil.: if the particle reaches b it must have 0 vel. and 0 force field hence the particle is fixed at b forever. What really happens is the particle never reaches b. Consider the time it takes for the particle to get from a to b $\Delta t(a\to b) = \sqrt{\frac{m}{2}}\lim_{\epsilon\to 0^+}\int_a^{b-\epsilon}\frac{dx}{\sqrt{E-U(x)}}$.

Now Taylor expand around $x = b$: $U(x) = U(b) + U'(b)(x-b) + \frac{1}{2}U''(b)(x-b)^2 + O((x-b)^3)$. If $U'(b) = 0$ and $E = U(b)$: $E - U(x) = -\frac{1}{2}U''(b)(b-x)^2 + O((x-b)^3)$. As b is a local max of the potential, $U''(x)<0$ and so the integral is approx (ignoring the factor of $m/2$): $2|U''(b)|^{-1/2}\int\frac{dx}{b-x} \approx -2|U''(b)|^{-1/2}\log(b-x)$ which is unbdd as $x\to b$. Even if $U''(b) = 0$ a similar argument shows unbddness. Thus, if b is a CP of the potential, the particle never reaches b.

4. The Two-Body Problem

For $x_1,x_2\in\mathbb{R}^3$ the positions of the two particles, masses m_1 and m_2 respectively; let $U(x_1,x_2) = U(|x_1-x_2|)$ be the potential of their interaction. Newt's eqn for the two body system is: $m_1\ddot{x}_1 = -\nabla_1 U, m_2\ddot{x}_2 = -\nabla_2 U$ where ∇_1 stands for the gradient operator wrt x_1, etc. Since the potential only depends on the distance between the particles, $U = U(|x_1-x_2|)$ it follows that $\nabla_1 U = -\nabla_2 U$,hence$m_1\ddot{x}_1 + m_2\ddot{x}_2 = 0$. This in turn implies that the qtty $p_c := m_1\dot{x}_1 + m_2\dot{x}_2$ is cnsrvd. This can be seen as the momentum of the center-of-mass (CoM). The centre-of-mass coordinate x_c for the two-body system as: $x_c := \frac{m_1x_1}{m_1+m_2} + \frac{m_2x_2}{m_1+m_2}$. Then $p_c = (m_1+m_2)\dot{x}_c$, so the cnsrvd qtty p_c is referred to as the centre-of-mass momentum of the two-body system.

Rewrite the problem in terms of the relative coordinate $x := x_1 - x_2$ and x_c s.t.: $\begin{pmatrix}x\\x_c\end{pmatrix} = \begin{pmatrix}1 & -1\\ \frac{m_1}{m_1+m_2} & \frac{m_2}{m_1+m_2}\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}$ $\begin{pmatrix}x_1\\x_2\end{pmatrix} = \begin{pmatrix}\frac{m_2}{m_1+m_2} & 1\\ -\frac{m_1}{m_1+m_2} & 1\end{pmatrix}\begin{pmatrix}x\\x_c\end{pmatrix}$.

Newt's eqns in these new coordinates is $\ddot{x}_c = 0$, $\mu\ddot{x} = -\nabla U(|x|)$, where ∇ denotes the gradient with respect to the relative coordinate x and $\mu := \frac{m_1m_2}{m_1+m_2}$ is called the **reduced mass** of the two body system. These are decoupled, and thus the two body system is equivalent to a system of two non-interacting particles. A free particle located at x_c of mass m_1+m_2, and a particle of mass μ located at x and acting under the conservative force field $-\nabla U$. The 1^{st} eqn of motion for the CoM can be solved as $x_c(t) = x_c(0) + \frac{p_ct}{m_1+m_2}$. The energy of the system then decomposes into the energies of each of the decoupled parts: $E = \frac{m_1}{2}|v_1|^2 + \frac{m_2}{1}|v_2|^2 + U(|x|) = \frac{m_1+m_2}{2}|\dot{x}_c|^2 + \frac{\mu}{2}|\dot{x}|^2 + U(|x|)$.

5. Elastic Collisions

When $U(|x|) = 0$ the two particles move freely, and $p_1 + p_2 = m_1v_1 + m_2v_2$ is cnsrvd in addition to the energy. In this scenario an **elastic collision** is when the two free particles collide and emerge from this collision with potentially different momenta. Using Galil. transform (motion with a constant vel.) to set $v_2 = 0$ the conservation of momentum and energy then give the relations: $p_1 = p_1' + p_2'$, $\frac{m_1}{2}|v_1|^2 = \frac{m_1}{2}|v_1'|^2 + \frac{m_2}{2}|v_2'|^2$. Where p_1',p_2',v_1' and v_2' are the respective quantities after the collision. Thus the motion takes place in the plane spanned by $\{p_1',p_2'\}$. Breaking the conservation of momentum into components parallel and perpendicular to p_1 gives: $m_1|v_1| = m_1|v_1'|\cos\theta_1 + m_1|v_2'|\cos\theta_2$, $0 = m_1|v_1'|\sin\theta_1 + m_2|v_2'|\sin\theta_2$, where θ_1,θ_2 are the angles between p_1',p_2' and p_1 respectively. (leaving 3 eqns to solve for 4 unknowns). s.t. can't model the collision, but can put constraints on the outgoing motion. This is done by finding constraints on θ_1. This is easiest relative to the CoM frame, where the CoM vel. is $v_c = \frac{m_1}{m_1+m_2}v_1$ is constant (as the particle motion is free). Then the relative velocities of each particle are $u_1 := v_1 - v_c$ and $u_2 := -v_c$ before the collision then $u_1' := v_1' - v_c$ and $u_2' := v_2' - v_c$ after the collision. Working in the CoM frame motion must be co-linear (otherwise CoM vel. would be 0). The conservation of energy gives $\frac{m_1}{2}|u_1|^2 + \frac{m_2}{2}|u_2|^2 = \frac{m_1}{2}|u_1'|^2 + \frac{m_2}{2}|u_2'|^2$. Subtracting the CoM energy from both sides and using the conservation of momentum gives $u_2 = -\frac{m_1}{m_2}u_1$ and $u_2' = -\frac{m_1}{m_2}u_1'$. Plugging this into the conservation of energy eqn gives $|u_1| = |u_1'|$ and so $|u_2| = |u_2'|$. So, relative to the CoM, the velocities are rotated by an angle, θ, which is not constrained by the conservation of momentum or energy.

Relate θ to θ_1 by decomposing $v_1' = u_1' + v_c$ into components parallel and perpendicular to v_c: $|v_1'|\cos\theta_1 = |u_1'|\cos\theta + |v_c|, |v_1'|\sin\theta_1 = |u_1'|\sin\theta$. Which can be rearranged to give: $\tan\theta_1 = \frac{\sin\theta}{\cos\theta + (m_1/m_2)}$.

5. Central Force Fields

Dfn: A **central force** is one that points in the direction of the line connecting the particle on which it acts with the origin for all configs.

Dfn: (Central Force Field). A force field $F:\mathbb{R}^3\to\mathbb{R}^3$ is called **central** if it takes the form $F(\mathbf{x}) = f(\mathbf{x})\mathbf{x}$, for some func $f:\mathbb{R}^3\to\mathbb{R}$.

Thm: (5.1). Let $F:\mathbb{R}^3\to\mathbb{R}$ be a conservative force field. Then F is central $\iff F(\mathbf{x}) = f(|F|)\mathbf{x} \iff F(\mathbf{x}) = -\nabla U(|\mathbf{x}|)$.

Dfn: For a particle of mass m the **ang mom** is $L = \begin{pmatrix}L_1\\L_2\\L_3\end{pmatrix} = \begin{pmatrix}x_2p_3 - x_3p_2\\ x_3p_1 - x_1p_3\\ x_1p_2 - x_2p_1\end{pmatrix}$. aka $L := \mathbf{x}\times p$. The components of the ang mom have dim $[L] = ML^2T^{-1}$.

Thm: (5.2). In any central force field, ang mom is a cnsrvd qtty.

Cor: (5.1). Motion in any central force field is restricted to a plane in $\mathbb{R}^3$.

Some Conditions for only bound motion: $\lim_{r\to\infty}U_{\text{eff}}(r) = \infty$. Furthermore, by showing $\lim_{r\to 0^+}U_{\text{eff}} = \infty$ and U_{eff} is finite for all $r\in(0,\infty)$ gives us that the motion can stay $r\in(0,\infty)$, rather than getting stuck at $r = 0$. Conditions for only scattering motion: $\lim_{t\to\infty}U_{\text{eff}}$ is finite, no minima at finite r, something like a wall at $r = 0$, e.g. $\lim_{r\to 0^+}U_{\text{eff}} = \infty$.

1D Effective Description

Use a Galil. rotation to orient our axes so that motion in a central field takes place in the (x_1,x_2)-plane: $L = (0,0,m\ell)^T$ Working in polar coords $x_1 = r\cos\theta$, $x_2 = r\sin\theta$ for the plane of motion. It follows momentum of a body subject to a central force is: $p = m\dot{\mathbf{x}} = m\begin{pmatrix}\dot{r}\cos\theta - r\dot{\theta}\sin\theta\\ \dot{r}\sin\theta + r\dot{\theta}\cos\theta\\ 0\end{pmatrix}$.

Conservation of ang mom then implies: $L_3 = x_1p_2 - x_2p_1 = mr^2\dot{\theta}$ is a constant and in particular $\ell = r^2\dot{\theta}$ is a cnsrvd qtty.

Princ: (Kepler's Area Law). The areal vel. of a body moving in a central force field is constant.

Adapting motion in a central field in terms of polar coords gives: $\frac{dp}{dt} = m\begin{pmatrix}(\ddot{r} - \dot{\theta}^2r)\cos\theta - (\ddot{\theta}r + 2\dot{\theta}\dot{r})\sin\theta\\ (\ddot{r} - \dot{\theta}^2r)\sin\theta + (\ddot{\theta}r + 2\dot{\theta}\dot{r})\cos\theta\\ 0\end{pmatrix}$. Using conservation of ang mom, taking a time deriv of $\ell = r^2\dot{\theta}$ gives the identity $2\dot{r}\dot{\theta} + r\ddot{\theta} = 0$ which simplifies the motion to $\frac{dp}{dt} = m\left(\ddot{r} - \frac{\ell^2}{r^3}\right)(\cos\theta,\sin\theta,0)^T$ By Newt's 2^{nd} law $\dot{p} = F$ while Thm 5.1 implies $F(\mathbf{x}) = -\frac{U'(|\mathbf{x}|)}{|\mathbf{x}|}\mathbf{x} = -U'(r)(\cos\theta,\sin\theta,0)^T$.

So Newt's eqn for motion in a conservative central force field reduces to a 1D eqn: $m\ddot{r} = -U'(r) + \frac{m\ell^2}{r^3}$ or $m\ddot{r} = -U_{\text{eff}}'(r)$ where $U_{\text{eff}} := U(r) + \frac{m\ell^2}{2r^2}$, $U_{\text{eff}}(|\mathbf{x}|) = U(|\mathbf{x}|) + \frac{|L|^2}{2m|\mathbf{x}|^2}$. Conservation of energy for the 1D effective description of motion states $E = m\dot{r}^2/2 + U(r) + \frac{m\ell^2}{2r^2}$ is cnsrvd.

Note that $\frac{m\ell^2}{2r^2} = mr^2\dot{\theta}^2/2$ for centrifugal energy. NOTE that $E = U_{\text{eff}}$ are NOT CPs with 0 vel.; just local mins/max of $r(t)$, there can still be movement in the angular direction (ie possible for $\dot{\theta}\neq 0$). The energy eqn above is equal to the total energy of the original 2D eqn as $m|\dot{\mathbf{x}}|^2 = m\dot{r}^2/2 + mr^2\dot{\theta}^2$.

In the 1D effective description, there are two classes of motion: $r\in[r_{\min},\infty)$ (scattering), $r\in[r_{\min},r_{\max}]$ (bound). In scattering, the body approaches the origin from some initial (possibly infinite) radius, achieves a closest approach of $r_{\min}$ ($E = U_{\text{eff}}$) and then flies away to ∞. In bound, the body orbits the origin between the closest and furthest radii i.e. bound orbits lie in the annulus bdd by $[r_{\min},r_{\max}]$ in the plane. By conservation of energy and ang mom $\frac{d\theta}{dr} = \frac{\dot{\theta}}{\dot{r}} = \pm\ell\left(r^2\sqrt{(2/m)(E-U(r)) - \ell^2/(r^2)}\right)^{-1}$.

As here, Newt's eqs are invariant under time inversion, the angle through which $\mathbf{x}(t)$ passes as it goes from $r_{\min}$ to $r_{\max}$ is half of $\Delta\theta$ so $\Delta\theta = 2\ell\int_{r_{\min}}^{r_{\max}}\frac{dr}{r^2\sqrt{(2/m)(E-U(r)) - \frac{\ell^2}{r^2}}} = 2\ell\int_{r_{\min}}^{r_{\max}}\frac{dr}{r^2\sqrt{(2/m)(E-U_{\text{eff}}(r))}}$.

Lem: (5.1). The orbits of bound motion in a conservative central force is closed iff $\Delta\theta = 2\pi n/m$ where $n,m\in\mathbb{Z}$ are coprime.

6. The Kepler Problem

Dfn: The **Kepler problem** is the general problem of motion in conservative central forces with potentials of form $U(r) = -k/r$, where k is a physical constant, dim $= [k] = ML^3T^{-2}$. $k>0\Rightarrow$attractive potential, $k<0\Rightarrow$repulsive.

The 1D effective potential for the Kepler problem is $U_{\text{eff}}(r) = -\frac{k}{r} + \frac{m\ell^2}{2r^2}$, where $\ell = r^2\dot{\theta}$ is a constant by the conservation of ang mom.

This maps onto the 2-body problem by decoupling the (free) CoM motion and taking $r = |x_1 - x_2|$, $m = \mu$ (the reduced mass of the two-body problem).

When $k<0$ and the potential is repulsive, $U_{\text{eff}}(r)>0$, $U_{\text{eff}}'(r)<0$ for all r, so only scattering motion is possible, with the two bodies reaching a minimal separation of $r_{\min}$ determined by $E = U_{\text{eff}}(r_{\min})$.

When $k>0$ the potential is attractive and bound motion is possible for configurations where $E<0$. In this case the radial motion is bdd by the roots of $E = U_{\text{eff}}(r)$; this is a quadratic eqn in r^{-1}, giving $\frac{1}{r_{\min}} = \frac{k}{m\ell^2}\left(1+\sqrt{1-\frac{2|E|m\ell^2}{k^2}}\right)$ $\frac{1}{r_{\max}} = \frac{k}{m\ell^2}\left(1-\sqrt{1-\frac{2|E|m\ell^2}{k^2}}\right)$.

Lem: For $k>0$, $E<0$ and $\ell\neq 0$, the turning angle for bound orbits of the Kepler problem is $\Delta\theta = 2\pi$.

The min of the effective potential with $k>0$ corresponds to the single (finite) root of $U_{\text{eff}}'(r) = k/(r^2) - m\ell^2/(r^3) = 0 \implies r = m\ell^2/k$. The value of the effective potential at this radius is $U_{\text{eff}}^{\min} = -\frac{k^2}{2m\ell^2}$. When $E = U_{\text{eff}}^{\min}$, $r_{\min} = r_{\max} = m\ell^2/k$ so the bound orbit is circular. Then by conservation of ang mom, $\dot{\theta} = \frac{k^2}{m^2\ell^3}$, and the period of the orbit is $T = \frac{2\pi m^2\ell^3}{k^2}$.

Dfn: (Laplace-Runge-Lenz vector). $A := \dot{\mathbf{x}}\times L + U\mathbf{x}$.

7. Kepler's Laws of Planetary Motion

The Laplace-Runge-Lenz (L-R-L) vector is orthogonal to L ($A\cdot L = 0$), thus A always lies in the plane of motion for the Kepler problem.

Dfn: Define φ as the angle between $\mathbf{x}$ and A in the plane of motion, then $\mathbf{x}\cdot A = |\mathbf{x}||A|\cos\varphi$. By the dfn of the Laplace-Runge-Lenz vec: $\mathbf{x}\cdot A = \mathbf{x}\cdot(\dot{\mathbf{x}}\times L) + U|\mathbf{x}|^2$, and taking $U = -k/|\mathbf{x}|$ and using the identity $a\cdot(b\times c) = c\cdot(a\times b)$ gives $\mathbf{x}\cdot A = m\ell^2 - k|\mathbf{x}|$. Equating with the earlier expression gives $|\mathbf{x}|\left(1 + \frac{|A|}{k}\cos\varphi\right) = \frac{m\ell^2}{k}$, paramizing the radial motion $r = |\mathbf{x}|$ in terms of the angle φ between the position and L-R-L vectors.

Dfn: A **conic section** in $\mathbb{R}^3$ is a planar curve obtained by intersecting an affine plane with a right circular double cone.

If (r,φ) are polar coordinates on the plane, then the curve of a conic section is given by $r(1+\varepsilon\cos\varphi) = \lambda$, where the focus of the conic section is at $r = 0$, the param $\varepsilon\ge 0$ is called the **eccentricity** and $\lambda>0$ is the **semi-latus rectum**. The latter is the distance from the focus at $r = 0$ to the conic section in the direction of the perpendicular to the major axis.

Dfn: The named cases of conic sections are given by the eccentricity: Circle when $\varepsilon = 0$, Ellipse when $\varepsilon<1$ (closed curves); Parabola when $\varepsilon = 1$, Hyperbola when $\varepsilon>1$ (open curves).

Lem: Motion in the Kepler problem is given by conic sections in the plane perpendicular to L with eccentricity $\varepsilon = \frac{|A|}{k}$ and semi-latus rectum $\lambda = \frac{m\ell^2}{k}$,

Lem: (7.2). For the attractive Kepler problem ($k>0$), $E<0$ iff the eccentricity of the resulting conic section is $\varepsilon<1$.

Princ: Kepler's 1^{st} Law: Bounded orbits in the Kepler problem are ellipses with the origin as one of the foci. (The orbit of a planet is an ellipse with the Sun at one of the two foci.)

Princ: Kepler's 2^{nd} Law: The areal vel. of motion (change in area w.r.t time) in the Kepler problem is constant.

Princ: Kepler's 3^{rd} Law: The period of a bdd orbit in the Kepler problem is proportional to the $\frac{3}{2}$ power of the length of the semi-major axis of the corresponding ellipse.

The min and max distances between the two bodies can be characterised by the quantities a and b; the lengths of the semi-major (a) and semi-minor (b) axes of the ellipse. Area(ellipse)$= \int_0^T\frac{dA}{dt}(t)dt$, where $A(t)$ is the area swept out by the trajectory $(r(t),\theta(t))$ as a func of time. By Kepler's 2^{nd} law $\frac{dA}{dt}(t) = \frac{r^2(t)}{2}\dot{\theta}(t) = \ell$ is constant so Area(ellipse)$= \int_0^T\ell/2dt = \ell T/2$, or equivalently $T = 2\text{Area(ellipse)}/\ell$.

Note $2a = r_{\min} + r_{\max}$. In the polar description of an ellipse ($\varepsilon<1$), $r_{\min}$ occurs at $\varphi = 0$ and $r_{\max}$ occurs at $\varphi = \pi$, so $r_{\min} = \frac{\lambda}{1+\varepsilon}$ and$r_{\max} = \frac{\lambda}{1-\varepsilon}$. Thus the length of the semi-major axis relates to the eccentricity and semi-latus rectum by $a = \frac{\lambda}{1-\varepsilon^2}$, $\Leftrightarrow r(1+\varepsilon\cos\varphi) = a(1-\varepsilon^2)$. Likewise the length of the semi-minor axis b corresponds to the max value taken by $r\sin\varphi = \frac{a(1-\varepsilon^2)\sin\varphi}{1+\varepsilon\cos\varphi}$. Thus the length of the semi-minor axis is $b = \left.\frac{a(1-\varepsilon^2)\sin\varphi}{1+\varepsilon\cos\varphi}\right|_{\max} = a\sqrt{1-\varepsilon^2}$. Now express a and b in terms of the physical params of the bound orbit for $a = k/(2|E|)$, $b = \sqrt{(m\ell^2)/(2|E|)}$.

Lem: (7.3). The area of an ellipse with semi-major and semi-minor axes of lengths a and b, respectively, is Area(ellipse)$= \pi ab$.

Substituting into T gives $T = \frac{2\pi ab}{\ell} = \frac{2\pi k\sqrt{m}}{(2|E|)^{3/2}}$, and our identity for b gives $T = 2\pi a^{3/2}\sqrt{m/k}$, in accordance with Kepler's 3^{rd} Law.

8. Scattering in Central Force Fields

Parabolic orbits ($\varepsilon = 0$) are only possible in the attractive ($k>0$) Kepler problem, for configurations with $E = 0$. The min radius $r_{\min}$ relative to the focus at the origin corresponds to $U_{\text{eff}}(r_{\min}) = 0$; so $r_{\min}^{\text{para}} = \frac{m\ell^2}{2k}$. WLOG paramise time using a Galil. transform s.t. $r(0) = r_{\min}$, which means that $\varphi(0) = 0$. Then $t = \frac{m^2\ell^3}{4k^2}\left[\tan\frac{\varphi}{2} + \frac{1}{3}\tan^3\left(\frac{\varphi}{2}\right)\right]$.

Dfn: For the cnsrvd quantities (E,ℓ) and geometric params (ε,λ) of a conic section: $\varepsilon = \sqrt{1+\frac{2m\ell^2}{k^2}E}$, $\lambda = \frac{m\ell^2}{k}$, $E = \frac{k(\varepsilon^2-1)}{2\lambda}$, $\ell = \sqrt{\frac{k\lambda}{m}}$.

Hyperbolic orbits, for both attractive and repulsive potentials of the Kepler problem, correspond to $E>0$ configurations. As $\lim_{r\to\infty}U_{\text{eff}}(r) = 0$;

Dfn: The **asymptotic speed**, v_∞, is the positive real number s.t $E = \frac{m}{2}v_\infty^2$ (since all energy at $r\to\infty$ has become kinetic).

Dfn: The **impact param** b associated with a given hyperbola is the min distance between either of its linear asymptotes and the origin.

Now add two more dictionaries. Firstly, between (E,ℓ) and (v_∞,b): $v_\infty = \sqrt{\frac{2E}{M}}$, $b = \sqrt{\frac{m\ell^2}{2E}}$, $E = \frac{M}{2}v_\infty^2$, $\ell = bv_\infty$, .Secondly, between $(\varepsilon,\ \ell)$ and $(v_\infty,\ b)$: $\varepsilon = \sqrt{1+\frac{m^2b^2v_\infty^4}{k^2}}$, $\lambda = \frac{mb^2v_\infty^2}{k}$, $v_\infty = \sqrt{\frac{k}{\lambda m}(\varepsilon^2-1)}$, $b = \frac{\lambda}{\sqrt{\varepsilon^2-1}}$.

Dfn: In a hyperbol. orbit, the closest $r_{\min}$ to the origin is $r_{\min} = -\frac{k}{mv_\infty^2} + \sqrt{b^2 + \frac{k^2}{m^2v_\infty^4}}$

As the body moves from ∞ to $r_{\min}$, the orbit passes through a turning angle χ before going back out to ∞. The **scattering angle** ψ is the angle of deflection the orbit experiences between its initial and final asymptotic trajectories; it is related to the turning angle by $\psi = |\pi - 2\chi|$.

Dfn: χ can be written $\chi = \ell\int_{r_{\min}}^\infty\frac{dr}{r^2\sqrt{\frac{2}{m}(E-U(r)) - \frac{\ell^2}{r^2}}}$. This can be substituted into $\psi = |\pi - 2\chi|$ for the scattering angle.

Lem: For hyperbol. scattering orbits in the Kepler problem, the scattering angle is $\psi = 2\arcsin\left(1/\left(\sqrt{1+\frac{m^2b^2v_\infty^4}{k^2}}\right)\right)$.

9. Small Oscillations and Stable Equilibria

Consider in 1D where Newt's eqn is $m\ddot{x} = -U'(x)$. If U has a CP use a Galil. translation to set the CP to be at $x = 0$, so that $U'(0) = 0$. Along with the assumption that $U(0) = 0$ (since only defined to a constant) means the Taylor expansion around the CP is $U(x) = \frac{U''(0)}{2}x^2 + O(x^3)$. Let $k := U''(0)$, then if $k = 0$ the cp is degen., stability is indeterminate. If $k>0$ then the cp is a stable (local min) and if $k<0$ it is unstable (local max).

Princ: (9.1). Simple harmonic motion, or Hooke's law, provides a universal 1^{st}-approx to motion near any stable equil. in a 1D mechanical system.

Dfn: (Radial Evolution Eq). For small perturbs for circular orbit, $r = r_0 + \rho$ for $\rho\ll 1$ — the radial evolution eqn is $m\ddot{\rho} = -U_{\text{eff}}''(r_0)\rho$. This corresponds to SHM with freq $\omega = \sqrt{U_{\text{eff}}''(r_0)/m}$.

Dfn: The **Hessian matrix** H is defined entry-wise by $H_{ij}(\mathbf{x}) = \frac{\partial^2 U}{\partial q_i\partial q_j}(\mathbf{x})$.

Thm: (Spectral thm for symmetric matrices) Let H be a real-valued 3×3 symmetric matrix with eigvals $k_i\in\mathbb{R}$ (for $i = 1,2,3$) — not necessarily distinct — and corresponding non-zero, lin. ind. eigvecs $v_i\in\mathbb{R}^3$, obeying $Hv_i = k_iv_i$. After normalising the eigvecs so that $|v_i| = 1\forall i$ and $v_i\cdot v_j = 0$ for $i\neq j$, the matrix $S = [v_1v_2v_3]$ is orthogonal ($S^TS = I$) and $HS = SD, D = \text{diag}(k_1,k_2,k_3)$.

Thm: (9.1). Motion near any stable equil. point is equivalent to three decoupled harmonic oscillators with eqns $m\ddot y = -Dy \Leftrightarrow \ddot y_i = -\frac{k_i}{m} y_i, i=1,2,3$. Specifically, the resulting solns are $y_i(t) = A_i \sin(\omega_i t + \varphi_i), \omega_i^2 := \frac{k_i}{m}$ with A_i and ω_i fixed as constants. The physical parametrisation of the mechanical system is obtained by $\mathbf{x}(t) = Sy(t)$.
Dfn: The variables $y_i(t)$ describing oscillatory motion around a stable equil. are called the **normal modes** of the mechanical system, and the frequencies ω_i are called the **characteristic frequencies**.

10. Near-Equilibrium Dynamics

Suppose $q_0 \in \mathbb{R}^N$ is an equil. point of a general mechanical system whose configuration space is N-dim.al and is conservative (with a well-defined potential $U:\mathbb{R}^N \to \mathbb{R}$). Let $\{e_i\}_{i=1,\dots,N}$ be the usual orthonormal basis of $\mathbb{R}^N$, so that a generic vector $q\in\mathbb{R}^N$ can be expanded as $q = \sum_{i=1}^n q_i e_i$, where $(q_1,\dots,q_n)$ are the corresponding coordinates on the configuration space $\mathbb{R}^N$. Thus $\nabla U(q_0)=0$, and since potential is only defined up to constants, can set $U(q_0)=0$, in which case Taylor expanding the potential around this equil. gives $U(q) = \frac12 (q-q_0)\cdot H(q_0)(q-q_0) + O(|q-q_0|^3)$.
Dfn: Assume the kinetic energy of the system is quadratic in the velocities, and can be written in terms of a $N\times N$ symmetric, positive-definite mass matrix M, as: $T = \frac12 \dot x \cdot M\dot x = \frac12\sum_{i,j=1}^N M_{ij}\dot x_i \dot x_j$.
Princ: (10.1, Near-Equilibrium Normal Form). The dynamics of small displacements from equil. can be studied by:
1. diagonalizing and normalizing the mass matrix, then
2. diagonalizing the potential and solving for the normal modes and characteristic frequencies of the system.

General Form of Solutions. Let M be the mass matrix. Then by spectral thm there exists an orthonormal matrix S_1 (made out of the normalised eigenvectors of M) s.t. $M' = S_1^T M S_1$, where M' is diagonal with positive entries (M is positive definite). Let D_1 be s.t. $M' = D_1^2$, then $M = S_1 D_1^2 S_1^T = (S_1D_1)(S_1D_1)^T$.
Now, define $z := (S_1D_1)^T x = D_1 S_1^T x$, which is invertible as $x = S_1 D_1^{-1} z$. In terms of z the kinetic energy is $T = \frac12\sum_{i=1}^N \dot z_i^2 = \frac12|\dot z|^2$. Thus normalising the near equil. dynamics. In terms of these new variables the potential energy is given by $U = \frac12 z\cdot Kz, K = D_1^{-1}S_1^T H S_1 D_1^{-1}$, to leading order around the point of equil.. K is symmetric since H and D_1 are symmetric and S_1 is orthogonal. Thus by the spectral thm for symmetric matrices, there must exist another orthogonal matrix S_2 s.t. $D := S_2^T K S_2$ is diagonal. Define $y := S_2^T z$, under which the form of kinetic energy is preserved $T = \frac12|S_2\dot y|^2 = \frac12|\dot y|^2$, since S_2 is orthogonal, while the potential energy becomes $U = \frac12 y\cdot Dy$. In particular, the resulting linear eqns of motion of the system are $\ddot y = -Dy$, which are N-decoupled linear eqns of motion as D is diagonal. Let $D = \text{diag}(\lambda_1,\dots,\lambda_N)$, then the decoupled eqns of motion are $\ddot y_i = -\lambda_i y_i, i=1,\dots,N$. Each one falls into three cases:
1. $\lambda_i>0$: oscillatory solns with characteristic frequency $\omega_i = \sqrt{\lambda_i}$ and normal mode $y_i(t) = A_i\sin(\omega_i t+\varphi_i)$.
2. $\lambda_i = 0$: linear, or zero-mode, solns with characteristic frequency $\omega_i = 0$ and normal mode $y_i(t) = a_i + b_i t$.
3. $\lambda_i<0$: exponential solns with imaginary characteristic frequency and normal mode $y_i(t) = A_i\exp\left(\sqrt{|\lambda_i|}t\right) + B_i\exp\left(-\sqrt{|\lambda_i|}t\right)$.

Dfn: (10.1, Equilibrium stability). A point of equil. is:
- **Stable** if $\lambda_i>0$ for all $i=1,\dots,N$,
- **Semi-stable** if $\lambda_i \ge 0$ for all $i=1,\dots,N$ and
- **Unstable** if there is at least one $i\in\{1,\dots,N\}$ for which $\lambda_i<0$.

11. Variational Calculus

Dfn: (Functional). A functional is a map from a set of funcs to $\mathbb{R}$. In particular, consider k-times cont.ly diffable funcs mapping from some closed domain $\Omega\subseteq\mathbb{R}^m$ to $\mathbb{R}^n$; this set of funcs is $C^k(\Omega,\mathbb{R}^n)$. Consider functionals $S: C^k(\Omega,\mathbb{R}^n)\to\mathbb{R}$ of form $S[f] = \int_\Omega L(f(x),\nabla f(x),\dots,f^{(k)}(x);x)d\mu_\Omega$, where $f\in C^k(\Omega,\mathbb{R}^n)$, $L:\mathbb{R}^{m+n\frac{m^{k+1}-1}{m-1}}\to\mathbb{R}$ is a smooth, real valued func of its arguments and $d\mu_\Omega$ is an integral measure on Ω.
Dfn: (Trajectories) Let $x:[0,1]\to\mathbb{R}^n$ be a trajectory with fixed boundary conditions $x(0)=P$, $x(1)=Q$ for 2 points $P,Q\in\mathbb{R}^n$. $C_{P,Q}$ is the space of all such trajectories.
Dfn: (endpoint fixed variations). Let $x(t),y(t)$ be any two paths in $C_{P,Q}$ and define $\varepsilon(t) := x(t)-y(t)$. As such, $\varepsilon:[0,1]\to\mathbb{R}^2$, $\varepsilon(0)=\varepsilon(1)=0$. Such a $\varepsilon(t)$ is called an **endpoint fixed variation**: it represents a shift in the initial path $x(t)$ which preserves the endpoints.
A trajectory x is an extremum of a functional $S[x]$ iff $\frac{d}{ds}S[x+s\varepsilon]\big|_{s=0}=0$, for all endpoint fixed variations $\varepsilon:[0,1]\to\mathbb{R}^n$ with $\varepsilon(0)=\varepsilon(1)=0$.
Thm: (Fundamental lemma of variational calculus). Let $f:[0,1]\to\mathbb{R}^n$ be a cont. func with $\int_0^1 f(t)\cdot h(t)=0$, for all $h\in C^\infty([0,1],\mathbb{R}^n)$ with $h(0)=h(1)=0$. Then $f(t)\equiv 0$.

12. Euler-Lagrange Eqns

Dfn: A **Lagrangian** is a smooth func $L:\mathbb{R}^{2n+1}\to\mathbb{R}$, sending $(x,\dot x,t)\mapsto L(x,\dot x;t)$. The **action** associated to L is the functional $S[x] = \int_0^1 L(x,\dot x;t)$ which is a map $S: C_{P,Q}\to\mathbb{R}$.
Consider the Lagrangian given by $L = |\dot x(t)|$. The action associated to this Lagrangian is the arclength of the path x from P to Q.
Thm: (12.1). A curve $x:[0,1]\to\mathbb{R}^n$ is an extremum of the action functional iff the E-L eqns $\frac{\partial L}{\partial x} = \frac{d}{dt}\frac{\partial L}{\partial\dot x} \Leftrightarrow \frac{\partial L}{\partial x_i} = \frac{d}{dt}\frac{\partial L}{\partial \dot x_i}$. Useful from Proof of Thm 12.1: $\frac{d}{ds}S[x+s\varepsilon]|_{s=0} = \int_0^1\frac{d}{ds}L(x,\dot x;t)|_{s=0}dt = \int_0^1\left(\frac{\partial L}{\partial x}\cdot\varepsilon + \frac{\partial L}{\partial\dot x}\cdot\dot\varepsilon\right)dt$.
Lem: (12.1). If $\frac{\partial L}{\partial x_i}=0$ for some $i\in 1,\dots,N$, then the generalised momentum component p_i is a cnsrvd qtty along curves obeying the E-L eqns.
Prop: (12.1). If the Lagrangian does not explicitly depend on time, $\frac{\partial L}{\partial t}=0$, then the energy is a cnsrvd qtty along curves obeying the E-L eqns.
Princ: (Hypersurfaces). Generalise the framework of extremising actions with fixed B.C.s, and let $x(0),x(1)$ be constrained to lie on hypersurfaces in $\mathbb{R}^n$, rather than points. Define the hypersurfaces $H_0,H_1\subset\mathbb{R}^n$ by $H_0 = \{x\in\mathbb{R}^n|g_0(x)=0\}, H_1=\{x\in\mathbb{R}^n|g_1(x)=0\}$ where $g_0,g_1:\mathbb{R}^n\to\mathbb{R}$ are smooth funcs. s.t. extremise the action subject only to $x(0)\in H_0$ and $x(1)\in H_1$.
Thm: Let $x:[0,1]\to\mathbb{R}^n$ be subject to $x(0)\in H_0$ and $x(1)\in H_1$ for hypersurfaces $H_0,H_1\subset\mathbb{R}^n$. Then x is an extremum of the action iff the E-L eqns are satisfied and the generalised momentum $p(t)$ obeys: $p(0)$ is normal to H_0 at $x(0)$ and $p(1)$ is normal to H_1 at $x(1)$.
Ex: Using the arclength functional find the shortest path from (2,1) ending on the hyperbola $xy=1$. SOL: By earlier examples on the arclength functional, the soln must be a straight line. Furthermore by the theory of general variations these extrema must be sols to the EL eqns and normal to the boundary hyper-surface at the point of intersection. So seek a line of form $x(t) = (1-t)\binom{2}{1} + t\binom{x_1}{y_1}$, which is normal to the parabola at (x_1,y_1). Here $g(x,y)=xy-1$, thus $\nabla g(x_1,y_1) = \binom{y_1}{x_1}$. Also, $\dot x = \binom{x_1-2}{y_1-1}$. Thus the soln being normal to the hyperbola is equivalent to the determinant of the matrix of $\nabla g(x_1,y_1)$ and $\dot x$ being equal to zero. Solve the resulting eqn using the relations in g to give the solns for x_1 and y_1.

13. The 2nd Variation

Dfn: (Symmetric bi-linear forms and quadratic forms). A symmetric bi-linear form on $\mathbb{R}^n$ is a map $B:\mathbb{R}^n\times\mathbb{R}^n\to\mathbb{R}$ which is linear in each of its arguments and symmetric. In other words $\forall x,y\in\mathbb{R}^n$ and $\forall a,b\in\mathbb{R}$ $B(ax,by) = abB(x,y), B(x,y)=B(y,x)$. The bi-linear property means that such a B can be represented as an $n\times n$ matrix, while the symm property means this matrix is symmetric.
Dfn: (Positive/negative-definite). A symmetric bi-linear form is positive(negative)-definite if it obeys $B(x,x)>0$ ($B(x,x)<0$) for all $x\in\mathbb{R}^n\setminus\{0\}$.
Lem: (13.1). A symmetric bi-linear form is positive(negative)-definite iff all of its eigenvalues are positive (negative).
Dfn: For any symmetric bi-linear form B there is an associated quadratic form $Q:\mathbb{R}^n\to\mathbb{R}$, $Q(x) := B(x,x)$.
Dfn: For any quadratic form Q the associated symmetric bi-linear form is given by the **polarization identity**: $B(x,y) = \frac12(Q(x+y)-Q(x)-Q(y))$.
Dfn: A norm on $\mathbb{R}^n$ is a map $Q:\mathbb{R}^n\to\mathbb{R}$ which obeys $Q(x)\ge0$ for all $x\in\mathbb{R}^n$ with $Q(x)=0$ iff $x=0$.
Lem: (13.2). A symmetric bi-linear form is positive-definite iff its associated quadratic form defines a norm on $\mathbb{R}^n$.
Dfn: (2^{nd} Variation). The 2^{nd} variation of an action functional $S[x]$ is the quadratic form $Q_x^S(\varepsilon) := \frac{d^2}{ds^2}S[x+s\varepsilon]\Big|_{s=0}$, acting on the space of variations $\varepsilon:[0,1]\to\mathbb{R}^n$ with appropriate boundary conditions (e.g. endpoint fixed variations).
Making use of the action in terms of the Lagrangian and restricting our attention to endpoint fixed variations, $Q_x^S[\varepsilon] = \int_0^1\sum_{i,j=1}^n\left(\frac{\partial L}{\partial x_i x_j}\varepsilon_i\varepsilon_j + 2\frac{\partial L}{\partial\dot x_i x_j}\dot\varepsilon_i\varepsilon_j + \frac{\partial L}{\partial\dot x_i\dot x_j}\dot\varepsilon_i\dot\varepsilon_j\right)dt$. For $n=1$; $Q_x^S[\varepsilon] = \int_0^1\left(\frac{\partial^2 L}{\partial x^2}\varepsilon^2 + 2\frac{\partial L}{\partial(\dot x, x)}\varepsilon\dot\varepsilon + \frac{\partial^2 L}{\partial\dot x^2}\dot\varepsilon^2\right)dt$. Then the polarization identity (by IBP with $\varepsilon(0)=\varepsilon(1)=0$ and $\eta(0)=\eta(1)=0$ gives $H_x^S(\varepsilon,\eta) = \int_0^1\eta\left(\frac{\partial^2 L}{\partial x^2} - \frac{d}{dt}\left(\frac{\partial L}{\partial(\dot x, x)}\right) - \frac{d}{dt}\left(\frac{\partial L}{\partial\dot x}\right)\frac{d}{dt}\right)\varepsilon dt$. Which can be written compactly as $H_x^S(\varepsilon,\eta) = \int_0^1\eta J(\varepsilon)dt$, where $J = \frac{\partial^2 L}{\partial x^2} - \frac{d}{dt}\left(\frac{\partial L}{\partial x,\dot x}\right) - \frac{d}{dt}\left(\frac{\partial L}{\partial\dot x}\right)\frac{d}{dt}$, is a 2^{nd}-order differential operator in t. This is generalised into $n>1$ dims by $H_x^S(\varepsilon,\eta) = \int_0^1\left(\sum_{i,j=1}^n\eta_i J_{ij}(\varepsilon_j)\right)dt = \int_0^1\eta\cdot J(\varepsilon)dt$, where $J_{ij} := \frac{\partial L}{\partial x_i x_j} - \frac12\frac{d}{dt}\left(\frac{\partial L}{\partial\dot x_i x_j} + \frac{\partial L}{\partial x_i\dot x_j}\right) - \frac{d}{dt}\left(\frac{\partial L}{\partial\dot x_i\dot x_j}\frac{d}{dt}\right)$ define the components of the $n\times n$ matrix of 2^{nd}-order differential operators J. This matrix is called the **Jacobi operator** of the Lagrangian.
A soln x to the Euler-Lagrange eqns is a (local) min of the action $S[x]$ iff H_x^S is positive-definite on the space of endpoint fixed variations, or, equivalently iff $Q_x^S(\varepsilon)>0$ for all non-trivial endpoint fixed variations ε. Likewise, x is a (local) max iff H_x^S is negative-definite or if $Q_x^S(\varepsilon)<0$ for all non-trivial endpoint fixed variations.

14. Principle of Least Action

Princ: (Hamilton's Principle of Least Action). Recall that Newt's eqns for a conservative mechanical system of the general form is $M\ddot x = -\nabla U(x)$ where the dim of configuration space is N, M is the symmetric $N\times N$ mass matrix of the system and $U(x)$ is the potential. Hamilton's principle of least action states that these eqns always arise as the extrema of an action functional whose Lagrangian takes a particular form.
Thm: (14.1). Motion for a mechanical system coincides with the extremals of the action functional with Lagrangian $L(x,\dot x;t) = T-U$ where T is kinetic energy and U is potential.
Lem: (14.1). For the Lagrangian $L=T-U$ of a conservative mechanical system, the energy is $E=T+U$ agreeing with the physical definition of total energy of the mechanical system.
Ex: (Free Particle). Write the Lagrangian for a free particle of mass m and show that the solns of the E-L eqns minimize the action. For a free particle the only energy is kinetic energy so $U=0$ and $T=\frac{m}{2}|\dot x|^2, x:[0,1]\to\mathbb{R}^3$ with $L=T=\frac{m}{2}|\dot x|^2$. The E-L eqns are simply $\frac{d}{dt}=m\ddot x=0$ whose solns are straight lines. Note that since $\frac{\partial L}{\partial x}=0$ the momentum $p=\frac{\partial L}{\partial\dot x}=m\dot x$ is cnsrvd along with energy. Calculating the 2^{nd} variation of the action: $Q_x^S(\varepsilon) = m\int_0^1|\dot\varepsilon|^2dt>0$ for all nontrivial ε, so the solns of the E-L eqns are the minima of the action.
Ex: (Pendulum with a moving pivot). Consider a planar pendulum whose pivot is a body of mass M which can move freely in the horizontal direction. Write the Lagrangian and E-L eqns for this system.
Place the origin on the horizontal line along which the mass M can move, and let x denote the horizontal displacement of the mass M from this point. Let $x_2 = \binom{x+L\sin\theta}{-L\cos\theta}$ denote the position of the mass m at the end of the pendulum, relative to origin. Then $|\dot x_2|^2 = \dot x^2 + 2L\cos\theta\dot x\dot\theta + L^2\dot\theta^2$, so the kinetic energy is given by $T = \frac{M}{2}\dot x^2 + \frac{m}{2}|\dot x_2|^2 = \frac{(M+m)}{2}\dot x^2 + \frac{m}{2}L^2\dot\theta^2 + mL\cos\theta\dot x\dot\theta$. This gives the mass matrix as $\begin{pmatrix} M+m & mL\cos\theta\\ mL\cos\theta & mL^2\end{pmatrix}$ which is not diagonal! The only potential from the system is gravitational from the pendulum which is $-mgL(1-\cos\theta)$. The Lagrangian is thus $\mathcal{L} = \frac{M}{2}\dot x^2 + \frac{m}{2}|\dot x_2|^2 = \frac{(M+m)}{2}\dot x^2 + \frac{m}{2}L^2\dot\theta^2 + mL\cos\theta\dot x\dot\theta - mgL(1-\cos\theta)$. Note that as $\frac{\partial\mathcal{L}}{\partial x}=0$ the momentum $p=(M+m)\dot x + mL\cos\theta\dot\theta$ is cnsrvd, in addition to the total energy. The E-L eqns for the system are $\frac{d}{dt}\left(\dot x + \frac{m}{M}\frac{L\cos\theta\dot\theta}{1+\frac{m}{M}}\right)=0$, $-\sin\theta\left(\frac{g}{L}+\frac{\dot x\dot\theta}{L}\right) = \frac{d}{dt}\left(\dot\theta + \frac{\cos\theta\dot x}{L}\right)$.

15. Noether's Thm

Diffeomorphisms and symmetries:

Let $S[x]$ be a functional, assume $x\in C^1([0,1],\mathbb{R}^n)$ and $x(0)=P$ and $x(1)=Q$ for $P,Q\in\mathbb{R}^n$. Consider a map $\varphi:\mathbb{R}^n\to\mathbb{R}^n$ which is twice cont.ly diffable. It follows that $y := \varphi\circ x, y = \varphi(x(t))$ is also a member of $C^1([0,1],\mathbb{R}^n)$ with endpoints $y(0)=\varphi(P)$ and $y(1)=\varphi(Q)$ in $\mathbb{R}^n$.
Dfn: (C^2 Diffeomorphism). A map $\varphi:\mathbb{R}^n\to\mathbb{R}^n$ is called a C^2 diffeomorphism if φ is invertible and both φ and φ^{-1} are in $C^2(\mathbb{R}^n,\mathbb{R}^n)$.
Now let $L:\mathbb{R}^{2n+1}\to\mathbb{R}$ be any smooth Lagrangian and define $K:\mathbb{R}^{2n+1}\to\mathbb{R}$ by $K(x,\dot x;t) := L(\varphi(x), D\varphi(x)\dot x;t)$where $D\varphi(x) := \left(\frac{\partial\varphi}{\partial x_1}\cdots\frac{\partial\varphi}{\partial x_n}\right)$ is the Jacobian of partial derivatives of φ.
Lem: (15.1). Let $S[y]=\int_0^1 L(y,\dot y;t)dt$, $I[x] = \int_0^1 K(x,\dot x;t)dt$ then x extremises I iff $y=\varphi\circ x$ extremises S.
Dfn: (Symmetry and Invariance). If a C^2 diff.ism $\varphi:\mathbb{R}^n\to\mathbb{R}^n$ has $L(x,\dot x;t) = L(y,\dot y;t)$for $y=\varphi\circ x$., it is a **symmetry** of L/L is **invariant** under φ.
Cor: (15.1). If φ is a symm of the Lagrangian, then φ maps extrema of the action to extrema of the action.
Consider a 1 param family of maps $\{\varphi_s:\mathbb{R}^n\to\mathbb{R}^n|s\in\mathbb{R}\}$ subject to the conditions:
1. $\forall s\in\mathbb{R}$, φ_s is a C^2 func on $\mathbb{R}^n$,
2. $\varphi_s(x)$ is diffable in $s\in\mathbb{R}$,
3. $\varphi_0(x)=x$,
4. $\forall s,t\in\mathbb{R}$, $\varphi_s\circ\varphi_t = \varphi_{s+t}$.

These conditions ensure that φ_s is invertible with $\varphi_s^{-1}=\varphi_{-s}$. Thus the 1 param family is actually a 1 param group of C^2 diff.isms on $\mathbb{R}^n$.
Thm: (Noether's Thm). Let $S[x]=\int_0^1 L(x,\dot x;t)dt$ be an action for curves $x:[0,1]\to\mathbb{R}^n$ and let L be invariant under the one param group of C^2 diff.isms $\{\varphi_s\}$. Then the Noether charge $N(x,\dot x;t) := \frac{\partial L}{\partial\dot x}\cdot\frac{\partial\varphi_s}{\partial s}\Big|_{s=0}$ is a cnsrvd qtty along solns of the E-L eqns.
Cor: (15.2). Suppose a 1 param group of C^2 diff.isms $\{\varphi_s\}$ transform the Lagrangian s.t. $\frac{\partial L}{\partial s}(y,\dot y;t) = \frac{\partial K_s}{\partial t}(x,\dot x;t)$for $y=\varphi_s\circ x$ and some $K_s(x,\dot x;t)$. Then the Noether charge $N(x,\dot x;t) = \frac{\partial L}{\partial\dot x}\cdot\frac{\partial\varphi_s}{\partial s}\Big|_{s=0} - K_0(x,\dot x;t)$ is a cnsrvd qtty along solns of the E-L eqns.
Thm: (Noether's Thm – Infinitesimal). Let $\{\varphi_s\}$ be a one param family of C^2 diff.isms and let $\zeta(X) := \frac{\partial\varphi_s}{\partial s}\Big|_{s=0}$. If the Lagrangian L obeys $\frac{\partial L}{\partial x}\cdot\zeta + \frac{\partial L}{\partial\dot x}\cdot\dot\zeta = \frac{dK}{dt}$, then the Noether charge $N(x,\dot x;t) = \frac{\partial L}{\partial\dot x}\cdot\zeta - K$ is a cnsrvd qtty along solns of E-L eqns.

16. Generalising Noether's Thm + Exs

Dfn: (Non-autonomous system). A non-autonomous Lagrangian system is one for which the kinetic and potential energies have the functional dependencies $T=T(x,\dot x;t)$ and $U=U(x;t)$ respectively. Equivalently, the Lagrangian $L=T-U$ has explicit time dependence.
Let $\{\psi_s:\mathbb{R}^{n+1}\to\mathbb{R}^{n+1}\}$ be a one param family of C^2 diff.isms for $s\in\mathbb{R}$ which act as $\psi_s(x,t) = (y,\bar t)$subject to $y = x+s\zeta(x,t)+O(s^2)$, $\bar t = t + s\tau(x,t)+O(s^2)$for $\zeta := \frac{\partial y}{\partial s}\Big|_{s=0}$, $\tau := \frac{\partial\bar t}{\partial s}\Big|_{s=0}$. This expansion allows us to is off a variety of useful partial derivatives: $\frac{\partial y_j}{\partial x_k}\Big|_{s=0} = \delta_{jk}, \frac{\partial y_j}{\partial t}\Big|_{s=0}=0, \frac{\partial\bar t}{\partial x_k}\Big|_{s=0}=0, \frac{\partial\bar t}{\partial t}\Big|_{s=0}=1$ $\frac{\partial y_j}{\partial s, x_k}\Big|_{s=0} = \frac{\partial\zeta_j}{\partial x_k}, \frac{\partial y_j}{\partial s,t}\Big|_{s=0} = \frac{\partial\zeta_j}{\partial t}, \frac{\partial\bar t}{\partial s,x_k}\Big|_{s=0} = \frac{\partial\tau}{\partial x_k}, \frac{\partial\bar t}{\partial s,t}\Big|_{s=0} = \frac{\partial\tau}{\partial t}$. This allows us to define the notion of a symm for non-autonomous systems.
Dfn: A 1-param family of C^2 diff.isms $\{\psi_s(y,\bar t)\}$ is a symm of the Lagrangian if $L\left(y,\frac{dy}{d\bar t};\bar t\right)\frac{d\bar t}{dt} = L(x,\dot x;t)$.
Thm: (Noether's Thm for non-autonomous systems). Let the 1-param family of C^2 diff.isms $\{\psi_s = (y,\bar t)\}$ be an infinitesimal symm of the Lagrangian, in the sense that $\frac{\partial L}{\partial x}\cdot\zeta + \frac{\partial L}{\partial\dot x}\cdot\frac{d\zeta}{dt} = \frac{d}{dt}\left(\frac{\partial L}{\partial\dot x}\cdot\zeta\right) + \left(\frac{\partial L}{\partial x} - \frac{d}{dt}\frac{\partial L}{\partial\dot x}\right)\cdot\zeta$ holds. Then the Noether charge $N(x,\dot x;t) := L\tau + \frac{\partial L}{\partial\dot x}\cdot(\zeta - \dot x\tau)$ is a cnsrvd qtty along solns of the E-L eqns.
Lem: (16.1). A 1-param family of C^2 diff.isms is an infinitesimal symm of the non autonomous Lagrangian if $\frac{\partial L}{\partial t}\tau + \frac{\partial L}{\partial x}\cdot\zeta + \frac{\partial L}{\partial\dot x}\cdot\left(\frac{\partial\zeta}{\partial t} - \dot x\frac{\partial\tau}{\partial t}\right) + L\frac{\partial\tau}{\partial t}=0$.
Cor: (16.1). If a one-param family of C^2 diff.isms $\{\psi_s=(y,\bar t)\}$ act infinitesimally on the Lagrangian as $\frac{\partial L}{\partial t}\tau + \frac{\partial L}{\partial x}\cdot\zeta + \frac{\partial L}{\partial\dot x}\cdot\left(\frac{d\zeta}{dt} - \dot x\frac{d\tau}{dt}\right) + L\frac{d\tau}{dt} = \frac{dK}{dt}(x,\dot x;t)$, for some $K(x,\dot x;t)$, then the Noether charge $N(x,\dot x;t) := L\tau + \frac{\partial L}{\partial\dot x}\cdot(\zeta - \dot x\tau) - K$ is a cnsrvd qtty along solns of the E-L eqns.
Ex: (Linear Momentum). Consider the Lagrangian $L = \frac12\dot x\cdot M\dot x - U(x)$, where $x:[0,1]\to\mathbb{R}^n$ and M is the $n\times n$, real valued, symmetric mass matrix. Suppose that $U(x)$ does not depend on x_i for some $i=1,\dots,n$. Show that $\varphi_s(x)$, defined by $\varphi_s(x) = x_j + s$ if $j=i$ and $=x_j$ otherwise is a symm of the Lagrangian and deduce the associated cnsrvd qtty. As the Lagrangian is indep. of x_i and $\dot\varphi = \dot x$, it follows immediately that φ_s is a symm of the Lagrangian. It is also clearly a C^2 diff.ism and so by Noether's thm the Noether charge is $N = \frac{\partial L}{\partial\dot x}\cdot\frac{\partial\varphi_s}{\partial s}\Big|_{s=0} = \sum_{j,k=1}^n M_{jk}\dot x_k\delta_{ij} = M_{jk}\dot x_k$, which is the generalised momentum component p_i.

17. Hamiltonian Formalism

Dfn: (Regular Lagrangian). A Lagrangian $L(x,\dot x;t)$ is **regular**, or non-degen. if the $n\times n$ matrix with entries $\frac{\partial L}{\partial\dot x_i\partial\dot x_j}, i,j=1,\dots,n$, is invertible for all values of $(x,\dot x,t)\in\mathbb{R}^{2n+1}$.
Dfn: (Hamiltonian). Let $L(x,\dot x;t)$ be a regular Lagrangian. The func $H(x,p;t) := p\cdot\dot x - L(x,\dot x;t)$,for $\dot x = \dot x(x,p;t)$, is called the **Hamiltonian**.
Thm: (17.1, Hamilton's Eqns). The Euler-Lagrange eqns of a regular Lagrangian system are equivalent to the system of $2n$ 1^{st}-order differential eqns $\frac{dx}{dt} = \frac{\partial H}{\partial p}, \frac{dp}{dt} = -\frac{\partial H}{\partial x}$, where $H=H(x,p;t)$ is the Hamiltonian These eqns are called **Hamilton's Eqns** or the canonical form for the Euler-Lagrange eqns.
Dfn: (Phase Space). $\mathbb{R}^{2n}$ with coordinates (x,p) is the **phase space** of the mechanical system.
Upon changing variables from the state space $(x,\dot x)$ to the phase space (x,p) the total energy of a mechanical system is seen to be equal to the Hamiltonian: $E = \dot x\cdot\frac{\partial L}{\partial\dot x} - L = \dot x(x,p;t)\cdot p - L(x,\dot x;t) = H$.
Dfn: (Poisson bracket). Let $f,g:\mathbb{R}^{2n}\to\mathbb{R}$ be any two diffable funcs on the phase space. The **Poisson bracket** of f and g is defined by $\{f,g\} := \frac{\partial f}{\partial x}\cdot\frac{\partial g}{\partial p} - \frac{\partial f}{\partial p}\cdot\frac{\partial g}{\partial x}$.
Lem: (17.2). The time-evolution of any qtty $\Phi:\mathbb{R}^{2n}\to\mathbb{R}$ (that does not explicitly depend on time ($\frac{\partial\Phi}{\partial t}=0$)) on the phase space along solns to the eqns of motion is governed by the Poisson bracket with the Hamiltonian $\frac{d\Phi}{dt} = \{\Phi,H\}$. Furthermore, Φ is a cnsrvd qtty iff it 'Poisson commutes' with the Hamiltonian, $\{\Phi,H\}=0$.
Prop: (17.1). The Poisson bracket has the following properties:
- $\{\cdot,\cdot\}$ is bi-linear in its arguments. • $\{f,gh\} = \{f,g\}h + g\{f,h\}$.
- $\{f,g\} = -\{g,f\}$.
- Jacobi identity: $\{f,\{g,h\}\} = \{\{f,g\},h\} + \{g,\{f,h\}\}$

for all twice-diffable funcs $f,g,h:\mathbb{R}^{2n}\to\mathbb{R}$ on the phase space.
Lem: (17.3). If $\Phi_1(x,p)$ and $\Phi_2(x,p)$ are cnsrvd quantities, then so is $\{\Phi_1,\Phi_2\}$.
Thm: (17.2). Every cnsrvd qtty in a mechanical system corresponds to a one-param family of symmetries of that system.
This is not the exactly converse of Noether's thm(s), which require a one-param family of symmetries which are C^2 diff.isms. Whereas the proof above only implicitly assumes that the one-param family is cont.ly diffable

18. Canonical Transformations and Integrability

Dfn: (Canonical transform). Let (x,p) be Darboux coordinates on phase space. A change of variables $(x,p)\mapsto(X,P)$ is called a **canonical transform** if the Poisson brackets of (X,P), calculated as funcs of (x,p), obey $\{X_i,X_j\} = 0 = \{P_i,P_j\}, \{X_i,P_j\} = \delta_{ij}$.
Thm: Let (x,p) be Darboux coordinates on the phase space of a system subject to Hamilton's eqns with Hamiltonian $H(x,p,t)$. Let $x,p\mapsto(X,P)$ be a canonical transform and let $\tilde H$ be the Hamiltonian in the new variables Then the eqns of motion in the new variables are $\frac{dX}{dt} = \frac{\partial\tilde H}{\partial P}, \frac{dP}{dt} = -\frac{\partial\tilde H}{\partial X}$.
Φ_1 and Φ_2 are **independent** if the vectors $\nabla\Phi_1,\nabla\Phi_2\in\mathbb{R}^{2n}$ are lin. ind for all values of (x,p).
Dfn: (Integrable system). A mechanical system whose phase space is $2n$-dim.al is (Hamiltonian) **integrable** if it has n indep. cnsrvd quantities. $\Phi_1,\Phi_2,\dots,\Phi_n$, with $\Phi_1=H$ the Hamiltonian of the system, all of which Poisson commute: $\{\Phi_i,\Phi_j\}=0$,for all $i,j,\dots,n$.
Lem: Every 1D mechanical system is integrable.

19. Isoperimetric Problems

Lem: Let $x\in U\subseteq\mathbb{R}^n$ be an extremum of f subject to $g=0$. Then if $\nabla g(x)\ne0$, there exists a Lagrange multiplier $\lambda\in\mathbb{R}$ s.t. (x,λ) is a CP of the func $F:U\times\mathbb{R}\to\mathbb{R}$ given by $F:=f-\lambda g$.
Thm: Suppose that $x(t)\in C_{P,Q}$ be an extremum of S subject to $I[x]=0$ Then if $x(t)$ is not an extremum of I, there exists a Lagrange multiplier $\lambda\in\mathbb{R}$ s.t. $x(t)$ is an extremum of the functional $S[x]-\lambda I[x]$.
Dfn: (Method of Lagrange multipliers).
1. Ensure that $I[x]$ has no extrema obeying $I[x]=0$.
2. Solve the Euler-Lagrange eqns of $L-\lambda K$.
3. Fix constants of integration using the boundary conditions $x(0)=P$, $x(1)=Q$.
4. Fix the value of λ from $I[x]=0$.

This only gives a necessary condition for x to be a soln of the isoperimetric problem. Normally ok to manually check the resulting x is a soln. Just add a new Lagrange multiplier for each new constraint in the problem.

20. Holonomic/Nonholonomic Constraints

Dfn: (Constraints). To determine extrema of an action $S[x]$ for $x\in C_{P,Q}$ subject to a constraint $G(x,\dot x,t)=0$, note there are 4 different types:
1. Scleronomic; if G has no explicit time dependence.
2. Rheonomic if G depends explicitly on time.
3. Holonomic if G does not depend explicitly on $\dot x$.
4. Non-holonomic if G has explicit $\dot x$ dependence.

Let $S[x]$ be subject to a holonomic constraint $G(x,\dot x,t) = g(x,t)=0$ for $g:\mathbb{R}^n\times[0,1]\to\mathbb{R}$ smooth and $x\in C_{P,Q}$. The boundary conditions on x imply $g(P,0)=g(Q,0)$, we assume $\nabla g(x,t)\ne0\,\forall t\in[0,1]$.
Thm: (20.1). Let $x\in C_{P,Q}$ be a smooth extremum of $S[x]$ subject to the holonomic constraint $g(x,t)=0$, with $\nabla g(x,t)\ne0$ for all $t\in[0,1]$. Then there exists some $\lambda:[0,1]\to\mathbb{R}$ s.t. x obeys the E-L eqns of $L-\lambda g$.
This means the method of Lagrange multipliers now extends to the setting of holonomic constraints:
1. Establish that $\nabla g(x,t)\ne0$.
2. Solve the E-L eqns of $L-\lambda g$.
3. Fix constants of integration with boundary conditions.
4. Determine $\lambda(t)$ by imposing $g(x(t))=0$.

Dfn: A constraint (non-holo) $G(x,\dot x,t)=0$ is called a **Pfaffian constraint** if it is at most linear in $\dot x$.
Lem: (20.1). Let $x\in C_{P,Q}$ be a smooth extremum of $S[x]$ subject to a Pfaffian constraint $G(x,\dot x,t)=0$ with $\nabla G(x,\dot x,t)\ne0$ for all $t\in[0,1]$. Then there exists some $\lambda:[0,1]\to\mathbb{R}$ s.t. x obeys the E-L eqns $L-\lambda G$.